

Yokesh Ramasamy Bhalamuruhan

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Work Authorization: Eligible for EU Blue Card; willing to relocate

Portfolio: <https://yrb.web.app>
LinkedIn: <https://linkedin.com/in/yokeshrb>
GitHub: <https://github.com/Yokesh-R-B>

TECHNICAL SKILLS

Languages: Java, Python, SQL, C#
Frameworks: Spring Boot, FastAPI, React, LangChain
Backend: REST APIs, Hibernate, JPA
Databases: PostgreSQL, MySQL, MongoDB
Cloud & Infrastructure: Azure, Docker, Linux
Tools: Git, IntelliJ, Maven, CI/CD
Concepts: System Design, Distributed Systems, Microservices Architecture, Data Structures & Algorithms, Dependency Injection

EDUCATION

Bauhaus-Universität Weimar Germany
Master of Science in Digital Engineering (Grade: 1.6) 04/2023 – 09/2026
• Relevant Coursework: Machine Learning, Computer Vision, Simulation Methods, Stochastic Systems, Optimization, Data Structures & Algorithms, Object-Oriented Programming, Software Engineering

Anna University Chennai India
Bachelor of Engineering in Civil Engineering 04/2017 – 03/2021
• Co-author, *Int. J. Steel Struct.* (2024), DOI: 10.1007/s13296-024-00816-9

PROFESSIONAL EXPERIENCE

FORVIA Hella GmbH & Co. KGaA, Lippstadt Germany
Software Developer Intern / Master's Thesis 10/2024 – 09/2025
• Designed and built a distributed Python workflow engine to orchestrate automotive simulation jobs across HPC clusters
• Developed modular backend components for job scheduling, execution monitoring, and result aggregation
• Automated end-to-end simulation pipelines, eliminating 90% of manual processes and reducing overhead
• Integrated external APIs and data pipelines to connect simulation infrastructure with downstream systems
• Refactored legacy simulation scripts into structured backend services, improving scalability and reducing technical debt

Bauhaus-Universität Weimar Germany
Student Assistant 04/2024 – 09/2024
• Enhanced a Retrieval-Augmented Generation (RAG) system by integrating external knowledge sources and improving retrieval quality
• Supported evaluation and teaching activities at the Chair of Stochastics and Optimization

Tata Consultancy Services, Chennai India
Assistant System Engineer 04/2021 – 02/2023
• Developed and maintained Java Spring Boot microservices for a US-based financial services client, supporting high-volume transaction processing across distributed systems
• Implemented automated reconciliation pipelines to validate financial transactions against external records, identifying discrepancies and reducing manual exception handling
• Built backend APIs and batch processing workflows to handle large-scale financial data, ensuring accuracy and reliability across end-of-day settlement cycles

PROJECTS

RAG Chatbot Backend Application | *LangChain, FastAPI, Python, ChromaDB, LLM*
• Built a Retrieval-Augmented Generation (RAG) system enabling efficient querying of large document datasets
• Implemented backend APIs for document ingestion, vector indexing, and LLM-based response generation
• Designed conversational workflows with chat history storage using SQLite and support for follow-up queries

Order Management System | *Spring Boot, Java, PostgreSQL*
• Developed a backend system to manage users, products, and orders using REST APIs
• Designed relational data models with entities such as Order, Product, and User using Spring Data JPA
• Implemented order workflows including item management, total price calculation, and status tracking

COMPETITIONS

- Participated in TREC 2024 RAG Track, organized by National Institute of Standards and Technology (NIST), USA and University of Waterloo

LANGUAGES

German (B1/B2), English (C1), Tamil (Native)